RESTAURANT GROWTH POTENTIAL MODELING & STRATEGIC CLASSIFICATION SYSTEM

Internship Project Report Submitted by Shrushti Bhambare

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UNIFIED MENTOR PVT. LTD.
Gurugram,Haryana

CERTIFICATE

This is to certify that **Shrushti Bhambare** has successfully completed the project titled **"Restaurant Growth Potential Modeling & Strategic Classification System"** as part of the internship project work at Unified Mentor Pvt. Ltd.

The project focuses on restaurant business performance analysis, growth potential assessment, machine learning-based clustering, strategic classification, and interactive dashboard development.

The work demonstrates the practical application of Python, data analytics, machine learning, and data visualization techniques to support business decision-making.

Project Title: Restaurant Growth Potential Modeling & Strategic Classification System

Submitted by: Shrushti Bhambare

Organization: Unified Mentor Pvt. Ltd.

Acknowledgement

I sincerely express my gratitude to Unified Mentor Pvt. Ltd. for providing me with the opportunity to work on this project.

I would like to thank my mentors for their guidance, support, and valuable suggestions throughout the project. Their feedback helped me understand the practical applications of data analytics and machine learning in business decision-making.

I am also thankful to my college faculty members, friends, and family for their encouragement and support during the completion of this project.

Finally, I would like to acknowledge the developers of the Python libraries and tools used in this project, which helped make the analysis and dashboard development possible.

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Abstract

demand, delivery operations, and operating costs influence long-term growth. Evaluating restaurants using only revenue or order volume may not provide a complete understanding of their growth potential.

This project focuses on analyzing the SkyCity Auckland Restaurants & Bars dataset to assess restaurant performance and identify potential growth opportunities. The dataset includes information related to cuisine type, restaurant segment, location, growth factor, average order value, monthly orders, sales channels, revenue, operating costs, delivery expenses, and net profit.

Exploratory Data Analysis (EDA) is performed to understand business patterns and relationships between important variables. Feature engineering is used to create additional business indicators, followed by the calculation of an exploratory Growth Potential Index (GPI). Unsupervised machine learning techniques, including K-Means and Hierarchical Clustering, are applied to group restaurants according to similar business characteristics. Principal Component Analysis (PCA) is used for dimensionality reduction and visualization.

An interactive dashboard is developed using Streamlit and Plotly to display key performance indicators, financial comparisons, strategic classifications, growth rankings, and restaurant-level insights.

The project demonstrates how data analytics and machine learning can support structured restaurant performance assessment and strategic business decision-making.

Keywords: Restaurant Analytics, Growth Potential, Business Intelligence, Machine Learning, K-Means Clustering, Hierarchical Clustering, PCA, Streamlit, Strategic Classification.

Executive Summary

The project, "**Restaurant Growth Potential Modeling & Strategic Classification System,**" aims to analyze restaurant business performance and identify potential growth opportunities using data analytics and unsupervised machine learning.

The SkyCity Auckland Restaurants & Bars dataset is analyzed to understand revenue, net profit, order volume, average order value, sales-channel performance, operating expenses, delivery costs, and growth-related indicators.

The project begins with data cleaning and Exploratory Data Analysis to identify patterns and relationships within the dataset. Feature engineering is then performed to generate additional indicators such as Total Revenue, Total Net Profit, Profit Margin, Scale Indicator, Cost Burden Rate, and Revenue Quality.

An exploratory Growth Potential Index is developed using selected business indicators. Machine learning techniques, including K-Means and Hierarchical Clustering, are applied to identify groups of restaurants with similar characteristics. The resulting clusters are interpreted to develop strategic restaurant classifications.

The final Streamlit dashboard presents the results through interactive charts, filters, KPI cards, restaurant rankings, and business insights.

The completed dashboard reports the following key figures:

- Total Restaurants: 1,696
- Total Revenue: $77,739,306.80
- Total Net Profit: $7,866,306.13
- Average GPI Score: 50.82

The project provides a structured analytical approach for understanding restaurant performance and supporting business decisions. The GPI and strategic classifications are exploratory decision-support tools and are not validated predictions of future growth.

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Introduction

The restaurant industry is influenced by changing customer preferences, competition, operating expenses, delivery-platform usage, and differences in customer demand. Restaurant owners and managers need to understand not only their current financial performance but also the factors that may influence future growth.

Traditional performance evaluation often focuses on revenue, sales, and order volume. However, high revenue does not always indicate strong profitability or sustainable business growth. Factors such as operating expenses, delivery costs, commission rates, average order value, and sales-channel contribution must also be considered.

Data analytics provides an effective way to examine business information, identify patterns, and support decision-making. Machine learning techniques can further help group restaurants according to similarities in their operational and financial characteristics.

This project develops a Restaurant Growth Potential Modeling and Strategic Classification System using the SkyCity Auckland Restaurants & Bars dataset. It combines Exploratory Data Analysis, feature engineering, a Growth Potential Index, unsupervised machine learning, and an interactive Streamlit dashboard.

Project Overview

The project analyzes restaurant-level data to understand business performance and identify different restaurant profiles. It considers financial, operational, demand-related, and delivery-channel indicators.

The project includes the following major activities:

- Data collection and understanding.
- Data cleaning and preprocessing.
- Exploratory Data Analysis.
- Feature engineering.
- Growth Potential Index calculation.
- Machine learning-based clustering.
- Strategic classification.
- Business recommendations.
- Interactive dashboard development.

1.2 Scope of the Project

The scope of this project is limited to analyzing the provided restaurant dataset and developing an exploratory classification system. It focuses on comparing restaurant performance, identifying patterns, and presenting insights through an interactive dashboard.

The project does not claim to predict actual future restaurant growth. Instead, it provides a data-driven framework for comparing restaurant characteristics and supporting further business evaluation.

Problem Statement

Restaurant performance is commonly evaluated using individual indicators such as total revenue, monthly orders, or profit. However, these indicators alone may not provide a complete understanding of a restaurant's growth readiness.

A restaurant may generate high revenue but also have high operating expenses or delivery commissions. Another restaurant may have strong profitability but limited order volume or expansion opportunities.

There is a need for a structured system that combines financial, operational, demand-related, and delivery-channel indicators to evaluate restaurant performance more comprehensively.

The project addresses the following problems:

1. Lack of a combined approach to evaluate restaurant growth potential.
2. Difficulty in comparing restaurants using multiple business indicators.
3. Limited understanding of revenue and profitability across different sales channels.
4. Need to analyze operating costs and delivery-related expenses.
5. Difficulty in identifying groups of restaurants with similar business characteristics.
6. Need for an interactive dashboard to present restaurant performance and strategic insights.

Objectives

The main objectives of the project are:

1. To perform Exploratory Data Analysis on the restaurant dataset.
2. To analyze revenue, net profit, monthly orders, and average order value.
3. To study sales-channel contributions and delivery performance.
4. To examine operating expenses, commission rates, and delivery costs.
5. To create additional business indicators through feature engineering.
6. To calculate an exploratory Growth Potential Index.
7. To apply PCA and unsupervised machine learning techniques.
8. To identify restaurant groups using clustering algorithms.
9. To interpret clusters and develop strategic restaurant classifications.
10. To provide business insights and recommendations.
11. To develop an interactive dashboard using Streamlit and Plotly.

Literature Review

1 Overview

Business performance analysis uses financial and operational indicators to understand organizational performance. Revenue, profit margin, operating expenses, sales volume, and cost structure are commonly considered when evaluating business performance.

In the restaurant industry, customer demand, average order value, delivery channels, and operating costs also influence business performance. Analyzing these factors together can provide a broader understanding of restaurant operations.

2 Exploratory Data Analysis

Exploratory Data Analysis is used to understand the structure and characteristics of a dataset. It includes examining missing values, duplicates, numerical distributions, categorical features, and relationships between variables.

EDA helps identify patterns and unusual observations before applying machine learning techniques.

3 Business Performance and Profitability Analysis

Profitability analysis examines the relationship between revenue, costs, and net profit. It helps identify whether high sales are supported by healthy financial performance.

For restaurant businesses, channel-level revenue and costs are also important because in-store and delivery orders may have different operating and commission expenses.

4 K-Means Clustering

K-Means is an unsupervised machine learning algorithm that groups observations into a predefined number of clusters based on similarity.

It assigns each observation to the nearest cluster center and updates the centers iteratively. In this project, K-Means is used to identify groups of restaurants with similar business characteristics.

5 Hierarchical Clustering

Hierarchical Clustering groups observations by building a hierarchy of clusters. It helps examine similarities between restaurants and compare grouping patterns with K-Means results.

The Ward linkage method is used in the project.

6 Principal Component Analysis

Principal Component Analysis is a dimensionality reduction technique that transforms correlated numerical variables into a smaller set of components while retaining as much information as possible.

In this project, PCA is used to reduce feature dimensionality and support the visualization of restaurant clusters.

7 Research Gap

Existing individual performance measures may not fully represent restaurant growth readiness. Revenue, profitability, demand, operating costs, and delivery performance are often examined separately.

This project combines these indicators into an exploratory analytical workflow. It uses feature engineering, a Growth Potential Index, and unsupervised clustering to identify restaurant profiles and present the findings through an interactive dashboard.

Tools & Technologies

Tool / Technology	Purpose
Python	Programming and data analysis
Pandas	Data manipulation and cleaning
NumPy	Numerical calculations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Plotly	Interactive charts
Scikit-learn	Machine learning and preprocessing
Jupyter Notebook	EDA and model experimentation
Streamlit	Interactive dashboard development
Anaconda	Python environment management
Git	Version control
GitHub	Project repository hosting

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Dataset Description

The project uses the **SkyCity Auckland Restaurants & Bars** dataset.

The dataset contains restaurant-level information related to restaurant identity, cuisine, business segment, location, growth factor, average order value, monthly orders, revenue, costs, delivery operations, and profitability.

Dataset Attributes

Attribute Group	Important Columns	Description
Restaurant Information	RestaurantID, RestaurantName	Unique restaurant identification
Business Category	CuisineType, Segment	Restaurant type and business segment
Location	Subregion	Restaurant location
Growth and Demand	GrowthFactor, AOV, MonthlyOrders	Growth and demand indicators
Order Channels	InStoreOrdersCount, UberEatsOrdersCount, DoorDashOrdersCount, SelfDeliveryOrdersCount	Orders by channel
Revenue	InStoreRevenue, UberEatsRevenue, DoorDashRevenue, SelfDeliveryRevenue	Revenue by channel
Costs	COGSRate, OPEXRate, CommissionRate	Cost and commission indicators
Delivery	DeliveryRadiusKM, DeliveryCostOrder, SD_DeliveryTotalCost	Delivery-related information
Profitability	InStoreNetProfit, UberEatsNetProfit, DoorDashNetProfit, SelfDeliveryNetProfit	Channel-level net profit
Channel Shares	InStoreShare, UE_share, DD_share, SD_share	Share of business activity by channel

Important Features

GrowthFactor: A dataset-provided indicator associated with restaurant growth.

AOV: Represents the average order value.

MonthlyOrders: Represents the monthly order volume.

Revenue: Represents the income generated through different sales channels.

COGSRate: Represents the cost of goods sold rate.

OPEXRate: Represents the operating expense rate.

CommissionRate: Represents the commission-related cost indicator.

DeliveryRadiusKM: Represents the delivery coverage distance.

Net Profit: Represents profit generated through individual channels.

Data Cleaning & Validation

Data cleaning is an important step before performing analysis and machine learning. The dataset was inspected and prepared to improve consistency and usability.

The following activities were performed:

1. Loaded the CSV dataset using Pandas.
2. Inspected dataset dimensions and column names.
3. Checked data types and missing values.
4. Identified duplicate records.
5. Reviewed numerical and categorical features.
6. Examined descriptive statistics.
7. Prepared relevant features for analysis.
8. Handled missing values during machine learning preprocessing.
9. Applied numerical scaling and categorical encoding.

Numerical features were processed using median imputation and RobustScaler. Categorical features were processed using most-frequent imputation and OneHotEncoder.

Methodology

1.Project Workflow

Dataset Collection

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Data Understanding

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Data Cleaning and Validation

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Exploratory Data Analysis

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Feature Engineering

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Growth Potential Index Calculation

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Feature Preprocessing

↓

Principal Component Analysis

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Clustering Algorithms

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Cluster Profiling

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Strategic Classification

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Business Recommendations

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Streamlit Dashboard Development

2.Feature Engineering

Feature engineering involves creating new variables from existing dataset columns to improve business analysis.

The following indicators were developed:

Derived Feature	Description
TotalRevenue	Combined revenue across available channels
TotalNetProfit	Combined net profit across channels
ProfitMargin	Net profit relative to total revenue
ScaleIndicator	Combines monthly orders and growth factor
CostBurdenRate	Combines selected cost-rate indicators
AggregatorDependence	Combines selected third-party aggregator shares
DeliveryOrdersShare	Represents delivery-channel order share
RevenuePerOrder	Revenue relative to monthly orders
RevenueQuality	Combines AOV and profit margin

3.Growth Potential Index (GPI)

The Growth Potential Index is an exploratory scoring measure developed to compare restaurants using selected business indicators.

The project uses the following component weights:

Indicator	Weight	Treatment
Scale Indicator	0.25	Higher normalized value contributes positively
Cost Burden Rate	0.20	Treated inversely
Aggregator Dependence	0.20	Retained as a diagnostic indicator
Revenue Quality	0.20	Higher normalized value contributes positively

Delivery Expansion Indicator	0.15	Used as a delivery-related indicator

The weights total 1.00.

The index is used as an exploratory comparison measure. It depends on the selected features, normalization method, and project-defined weights. It is not a validated forecast of future restaurant growth.

4.GPI-Based Classification

The project uses the following exploratory score ranges:

GPI Score	Classification
70 and above	High Growth Potential
40 to below 70	Moderate Growth Potential
Below 40	Low Growth Potential

These ranges are project-defined thresholds and should be interpreted as analytical categories rather than verified business outcomes.

5.Machine Learning Approach

The project applies unsupervised machine learning to group restaurants according to their business characteristics.

The following techniques are used:

- **PCA:** Reduces dimensionality and supports visualization.
- **K-Means:** Groups restaurants based on similarity.
- **Hierarchical Clustering:** Creates an alternative grouping for comparison.
- **DBSCAN:** Explores density-based grouping and potential outliers.

The clustering process is unsupervised. It does not predict a known future growth label.

Exploratory Data Analysis

Dataset Preview

```
print("Dataset Preview")

display(df.head())
```

Dataset Preview

	CuisineType	RestaurantID	RestaurantName	Segment	Subregion	GrowthFactor	AOV	MonthlyOrders	InStoreOrders	InStoreRevenue	...	DeliveryCostPerOrder	SD
0	Burgers	25731	Urban Burgers House	Cafe	North Shore	1.03	43.97	668	197	8662.09	...	3.25	
1	Burgers	25123	Urban Burgers Diner	QSR	South Auckland	1.05	40.45	1388	259	10476.55	...	4.72	
2	Burgers	25177	King Burgers Eatery	Cafe	West Auckland	1.04	40.03	1717	524	20975.72	...	3.25	
3	Burgers	25540	Classic Burgers Tavern	QSR	North Shore	1.03	36.28	1083	216	7836.48	...	0.89	
4	Burgers	25258	Lucky Burgers Bistro	Cafe	South Auckland	1.05	34.34	1230	261	8962.74	...	2.66	

5 rows × 30 columns

Figure 1: Dataset Preview

The dataset was loaded and examined to understand its structure, features, and sample restaurant records.

Dataset Information

```
print("Dataset Columns:\n")

for col in df.columns:
    print(col)
```

```
Dataset Columns:

CuisineType
RestaurantID
RestaurantName
Segment
Subregion
GrowthFactor
AOV
MonthlyOrders
InStoreOrders
InStoreRevenue
UberEatsOrders
DoorDashOrders
SelfDeliveryOrders
UberEatsRevenue
DoorDashRevenue
SelfDeliveryRevenue
COGSRate
OPEXRate
CommissionRate
DeliveryRadiusKM
DeliveryCostPerOrder
SD_DeliveryTotalCost
InStoreNetProfit
UberEatsNetProfit
DoorDashNetProfit
SelfDeliveryNetProfit
InStoreShare
UE_share
DD_share
SD_share
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1696 entries, 0 to 1695
Data columns (total 30 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   CuisineType            1696 non-null   object
 1   RestaurantID           1696 non-null   int64
 2   RestaurantName         1696 non-null   object
 3   Segment                1696 non-null   object
 4   Subregion              1696 non-null   object
 5   GrowthFactor           1696 non-null   float64
 6   AOV                    1696 non-null   float64
 7   MonthlyOrders          1696 non-null   int64
 8   InStoreOrders          1696 non-null   int64
 9   InStoreRevenue         1696 non-null   float64
 10  UberEatsOrders         1696 non-null   int64
 11  DoorDashOrders         1696 non-null   int64
 12  SelfDeliveryOrders     1696 non-null   int64
 13  UberEatsRevenue        1696 non-null   float64
 14  DoorDashRevenue        1696 non-null   float64
 15  SelfDeliveryRevenue    1696 non-null   float64
 16  COGSRate               1696 non-null   float64
 17  OPEXRate               1696 non-null   float64
 18  CommissionRate         1696 non-null   float64
 19  DeliveryRadiusKM       1696 non-null   int64
 20  DeliveryCostPerOrder   1696 non-null   float64
 21  SD_DeliveryTotalCost   1696 non-null   float64
 22  InStoreNetProfit       1696 non-null   float64
 23  UberEatsNetProfit      1696 non-null   float64
 24  DoorDashNetProfit      1696 non-null   float64
 25  SelfDeliveryNetProfit  1696 non-null   float64
 26  InStoreShare           1696 non-null   float64
 27  UE_share               1696 non-null   float64
 28  DD_share               1696 non-null   float64
 29  SD_share               1696 non-null   float64
dtypes: float64(19), int64(7), object(4)
memory usage: 397.6+ KB
```

Figure 2: Dataset Information

Dataset dimensions, column names, and data types were inspected to understand the available information.

Missing Value Analysis

```
[6]: missing_data = missing_summary[
         missing_summary['Missing Values'] > 0
     ]

     if not missing_data.empty:

         plt.figure(figsize=(12, 6))

         sns.barplot(
             data=missing_data,
             x='Missing Percentage',
             y='Column'
         )

         plt.title('Missing Values by Column')
         plt.xlabel('Missing Values (%)')
         plt.ylabel('Columns')
         plt.tight_layout()
         plt.show()

     else:
         print("No missing values found.")
```

```
No missing values found.
```

Figure 3: Missing Value Analysis

Missing values were checked to identify incomplete information and determine the required preprocessing steps.

Descriptive Statistics of the Dataset

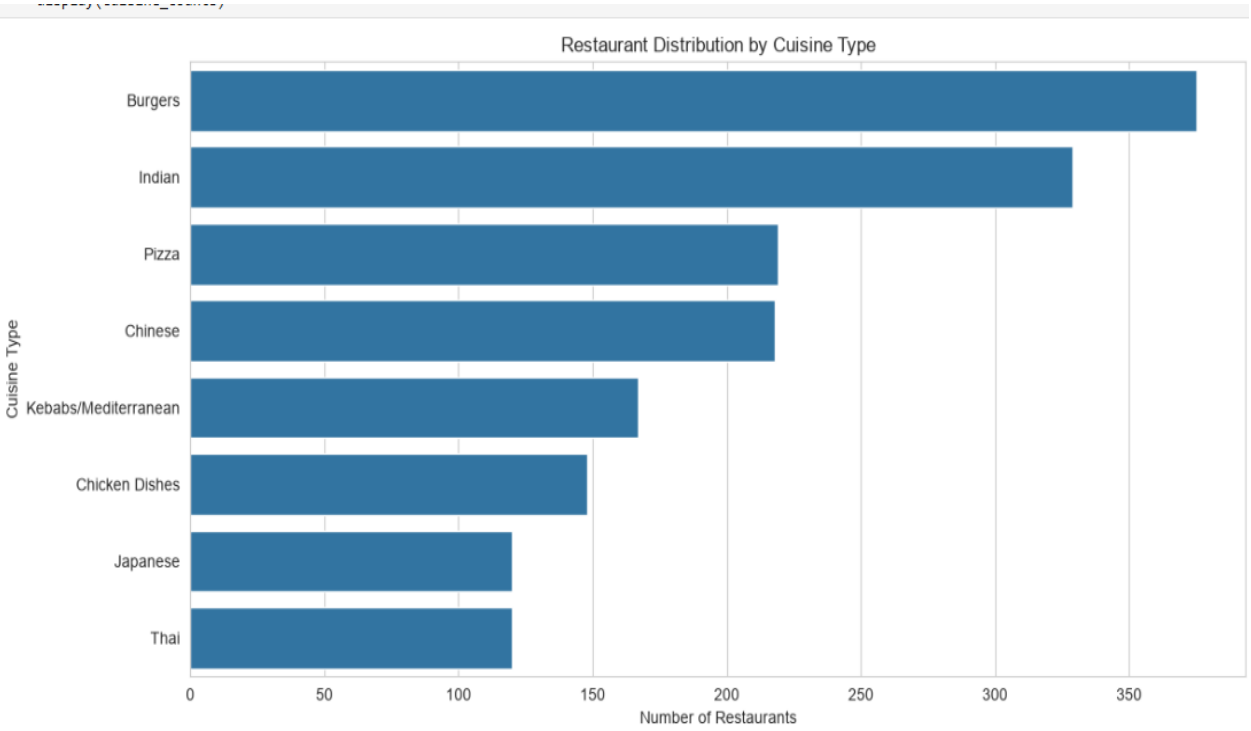
```
display(df.describe(include='all').T)
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
CuisineType	1696	8	Burgers	375	NaN	NaN	NaN	NaN	NaN	NaN	NaN
RestaurantID	1696.0	NaN	NaN	NaN	25848.5	489.737345	25001.0	25424.75	25848.5	26272.25	26696.0
RestaurantName	1696	256	Lotus Eatery	18	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Segment	1696	4	QSR	543	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Subregion	1696	4	West Auckland	462	NaN	NaN	NaN	NaN	NaN	NaN	NaN
GrowthFactor	1696.0	NaN	NaN	NaN	1.028255	0.0227	0.99	1.03	1.04	1.05	1.05
AOV	1696.0	NaN	NaN	NaN	38.516887	4.459408	29.79	34.7875	38.34	42.495	47.23
MonthlyOrders	1696.0	NaN	NaN	NaN	1190.527123	422.334848	441.0	827.0	1182.0	1525.0	2337.0
InStoreOrders	1696.0	NaN	NaN	NaN	218.980542	149.758619	13.0	101.0	178.0	307.0	819.0
InStoreRevenue	1696.0	NaN	NaN	NaN	8422.39273	5851.233352	412.36	3904.3425	6791.43	11550.755	35356.23
UberEatsOrders	1696.0	NaN	NaN	NaN	471.90625	171.781981	142.0	331.0	456.0	602.0	937.0
DoorDashOrders	1696.0	NaN	NaN	NaN	257.155071	92.605301	81.0	179.75	249.0	326.25	466.0
SelfDeliveryOrders	1696.0	NaN	NaN	NaN	242.485259	129.950736	60.0	146.0	212.0	306.25	702.0
UberEatsRevenue	1696.0	NaN	NaN	NaN	18170.03036	6959.241673	5050.95	12524.745	17548.98	23016.0	43038.96
DoorDashRevenue	1696.0	NaN	NaN	NaN	9899.399145	3750.087557	2876.7	6806.92	9564.8	12617.6025	21313.44
SelfDeliveryRevenue	1696.0	NaN	NaN	NaN	9345.03319	5191.322908	1897.2	5631.8525	8066.46	11852.9075	32481.54
COGSRate	1696.0	NaN	NaN	NaN	0.280149	0.06176	0.200369	0.23227	0.261271	0.352189	0.399826
OPEXRate	1696.0	NaN	NaN	NaN	0.41165	0.07342	0.200301	0.372182	0.411785	0.456814	0.549688
CommissionRate	1696.0	NaN	NaN	NaN	0.300277	0.017931	0.27	0.28	0.3	0.32	0.33

Figure 4: Descriptive Statistics of the Dataset

Numerical features were analyzed using descriptive statistics such as mean, standard deviation, minimum, maximum, and quartiles.

Categorical Feature Analysis

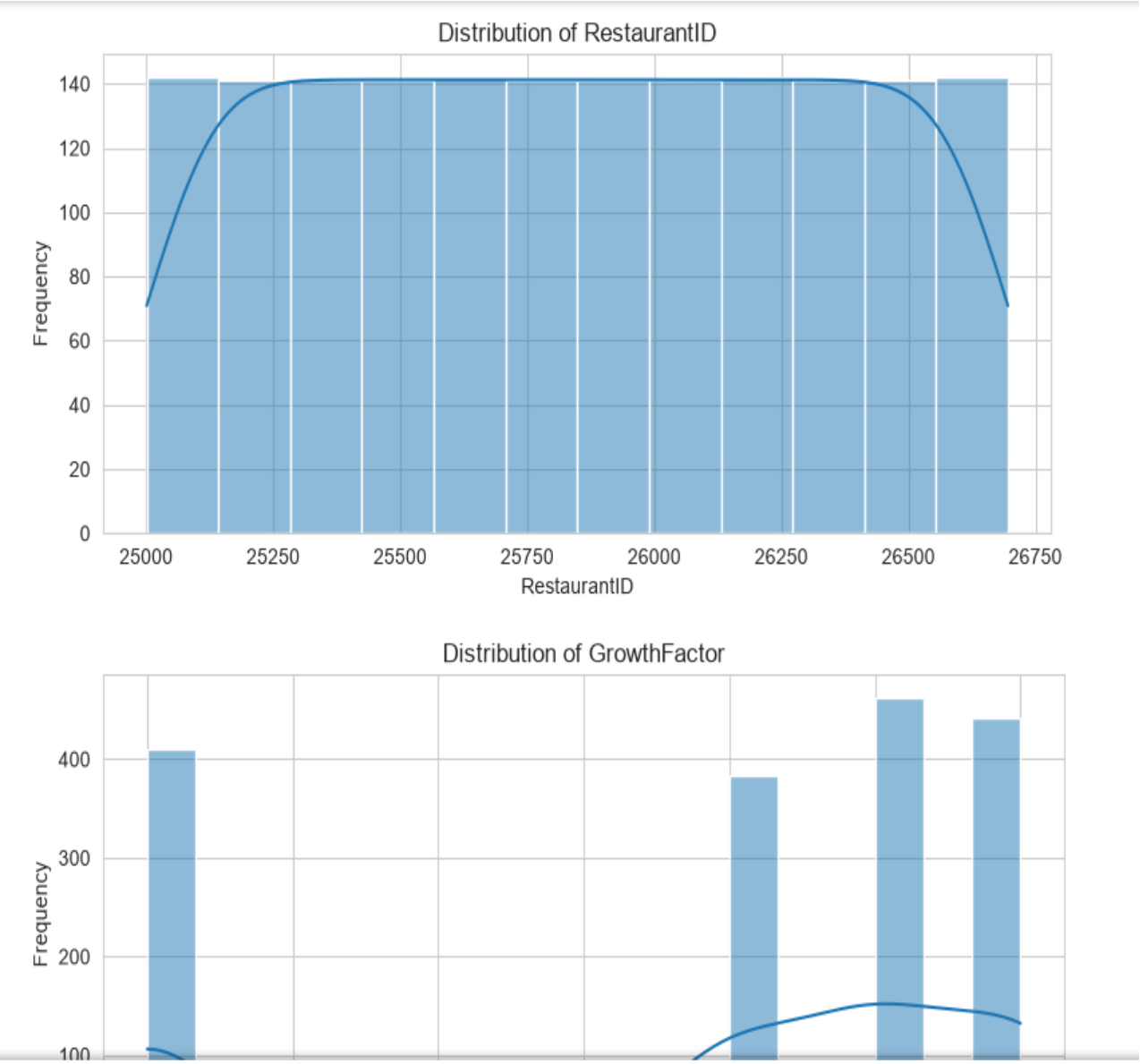



```
CuisineType
Burgers                375
Indian                 329
Pizza                  219
Chinese                218
Kebabs/Mediterranean   167
Chicken Dishes         148
Japanese               120
Thai                   120
Name: count, dtype: int64
```

Figure 5: Categorical Feature Analysis

Categorical variables such as cuisine type, restaurant segment, and subregion were examined to understand the distribution of restaurants.

Numerical Feature Analysis




```
for col in numerical_columns:
```

Figure 6: Numerical Feature Analysis

Numerical variables were explored to understand their distributions, ranges, and variations.

Restaurant Growth and Demand Analysis

```
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```

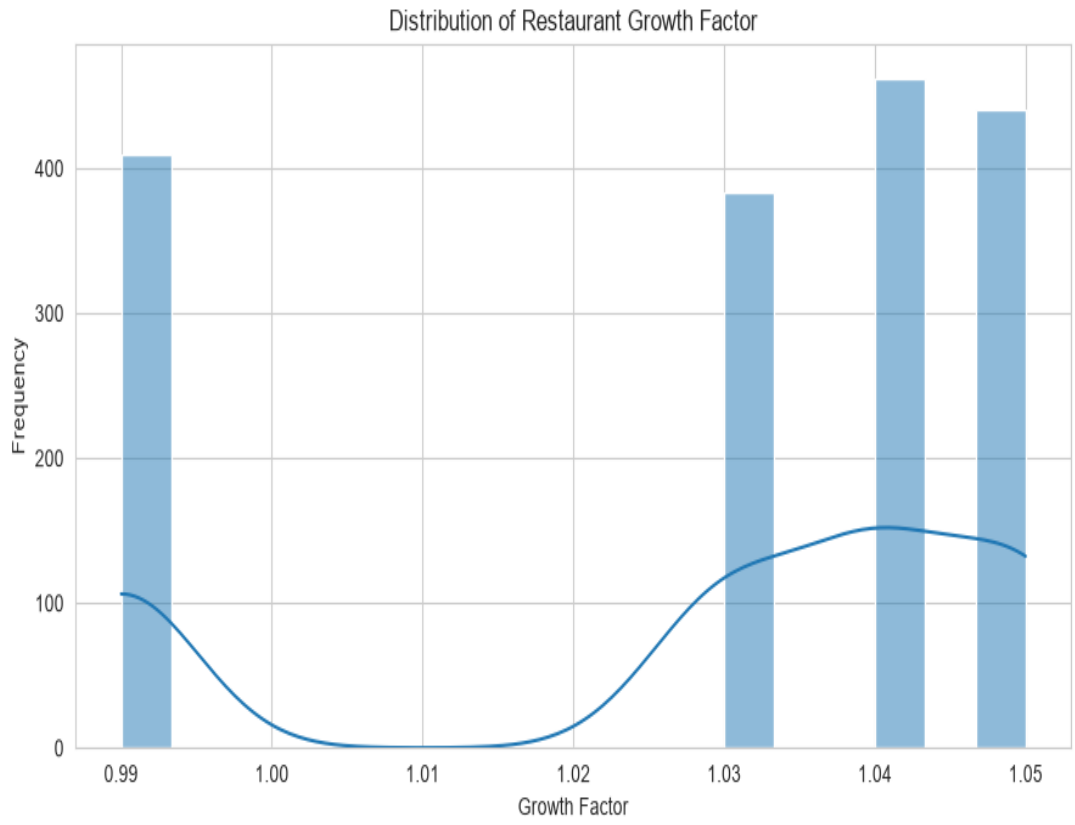
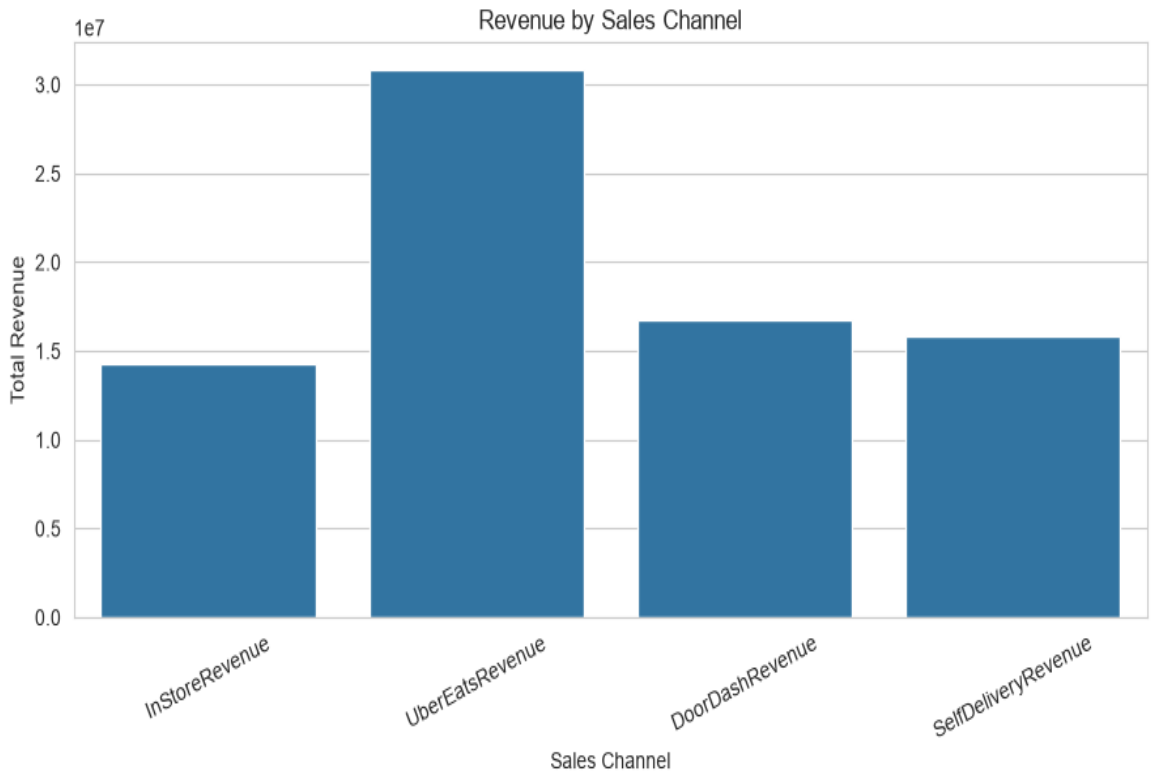


Figure 7: Restaurant Growth and Demand Analysis

GrowthFactor, AOV, and MonthlyOrders were analyzed to understand differences in restaurant demand and business scale.

Revenue and Sales Channel Analysis

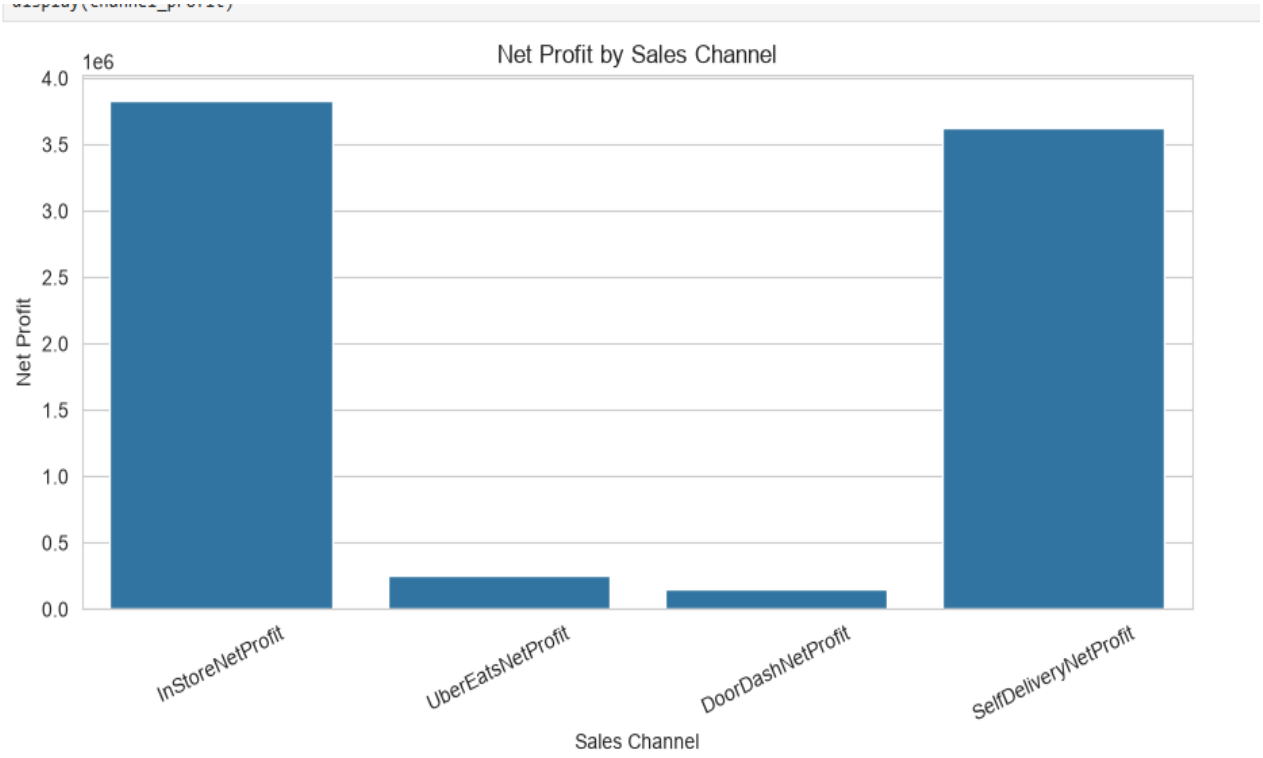



```
InStoreRevenue          14284378.07
UberEatsRevenue         30816371.49
DoorDashRevenue         16789380.95
SelfDeliveryRevenue     15849176.29
dtype: float64
```

Figure 8: Revenue by Sales Channel

Revenue from in-store, Uber Eats, DoorDash, and self-delivery channels was compared to understand the contribution of different sales channels.

Cost and Profitability Analysis




```
InStoreNetProfit         3828417.36
UberEatsNetProfit         258488.08
DoorDashNetProfit         151122.25
SelfDeliveryNetProfit    3628278.44
dtype: float64
```

Figure 9: Cost and Profitability Analysis

Cost-related indicators and channel-level net profit were examined to understand the financial performance of restaurants.

Delivery Channel and Logistics Analysis

```
plt.tight_layout()
plt.show()
```

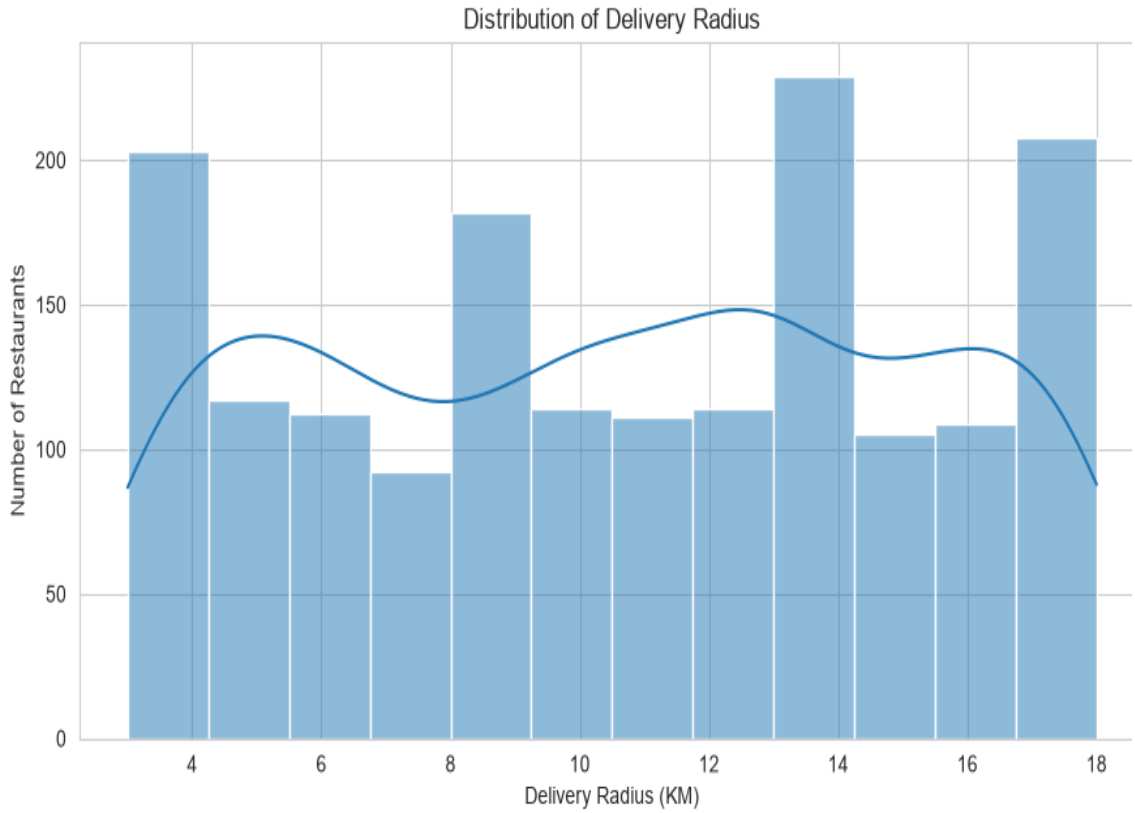


Figure 10: Delivery Channel Analysis

Delivery orders, delivery shares, delivery radius, and delivery costs were analyzed to understand delivery operations.

Correlation Analysis

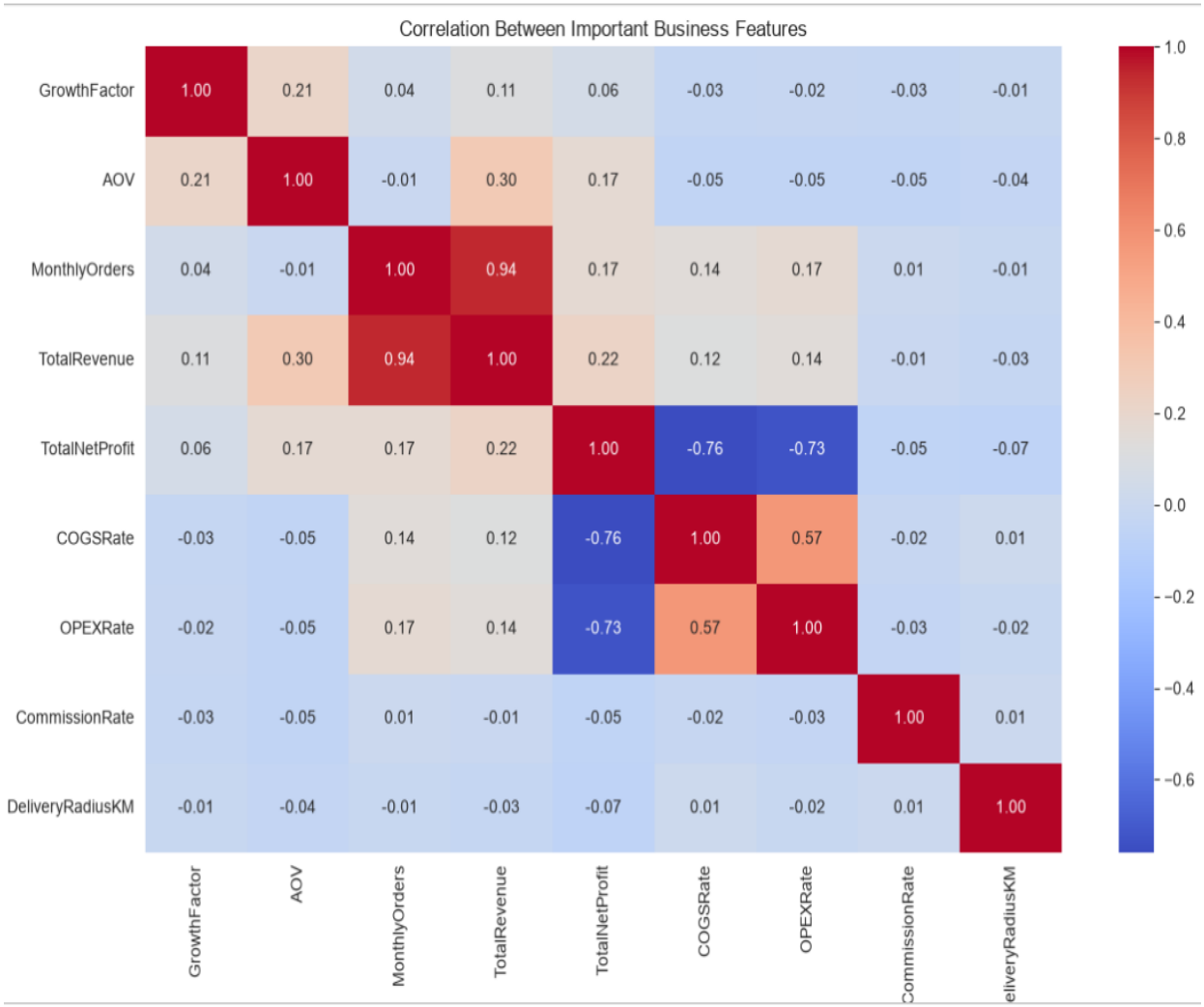


Figure 11: Correlation Heatmap

Correlation analysis was performed to examine relationships between selected numerical variables. A heatmap was used to visualize the correlation patterns.

Growth Potential Index Analysis

```
display(gpi_scaled.head())
```

	ScaleIndicator	CostBurdenRate	AggregatorDependence	RevenueQuality	DeliveryExpansionIndicator
0	0.124649	0.238585	0.5	0.833807	0.533333
1	0.506038	0.404886	0.5	0.581592	0.866667
2	0.668773	0.346909	0.5	0.674637	0.533333
3	0.336546	0.444504	0.5	0.515870	0.000000
4	0.423798	0.402964	0.5	0.520910	0.400000

Figure 12: Growth Potential Index Distribution

The Growth Potential Index was calculated to compare restaurants using selected business indicators. The resulting scores were used for ranking and exploratory readiness classification.

MACHINE LEARNING AND STRATEGIC CLASSIFICATION

1. Feature Selection

Relevant numerical and categorical features were selected for machine learning. Features were chosen based on their relevance to restaurant performance, demand, costs, profitability, and delivery operations.

2. Data Preprocessing

Numerical features were processed using median imputation and RobustScaler. Categorical features were processed using most-frequent imputation and one-hot encoding.

This prepared the dataset for machine learning algorithms.

3. Principal Component Analysis

PCA was applied to reduce the dimensionality of the prepared features. The workflow retained at least 90% cumulative explained variance.

PCA also supported the visualization of restaurant groups.

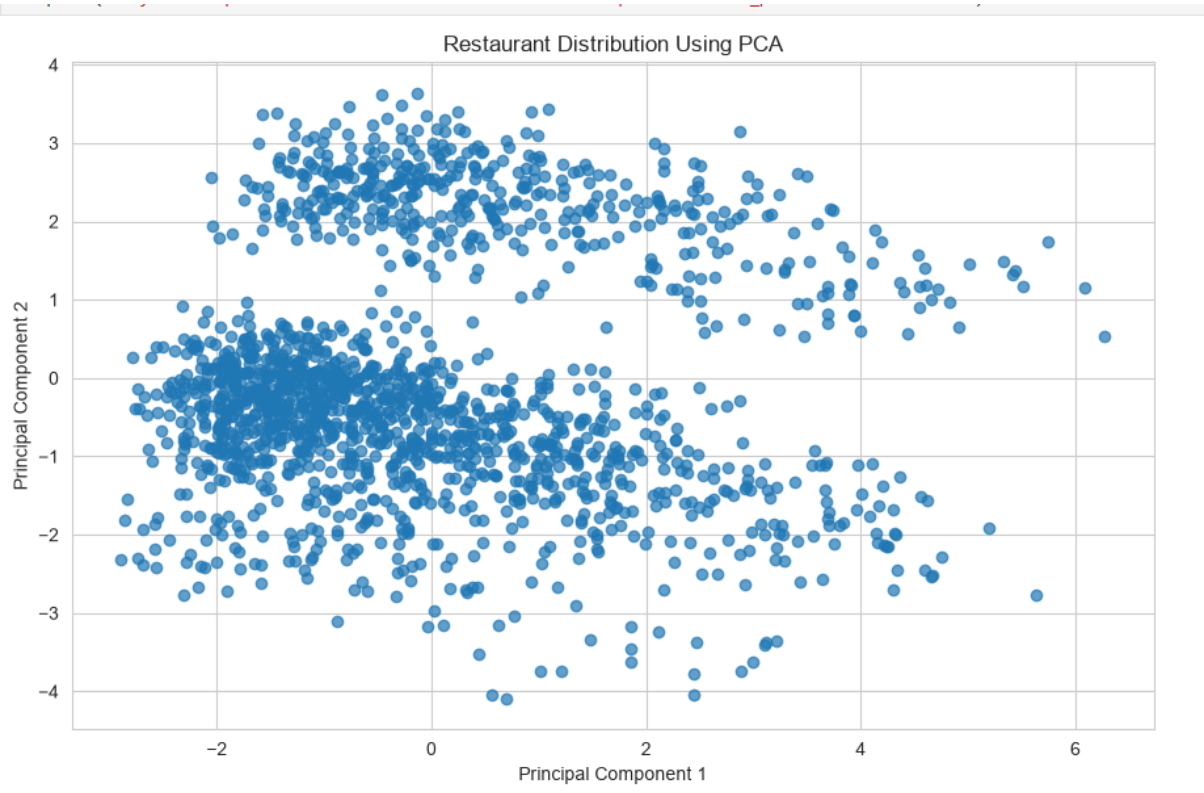


Figure 13: PCA Visualization

4. K-Means Clustering

K-Means clustering was applied to group restaurants based on similarities in their selected features. Candidate cluster counts were compared using silhouette scores to help select a suitable number of clusters.

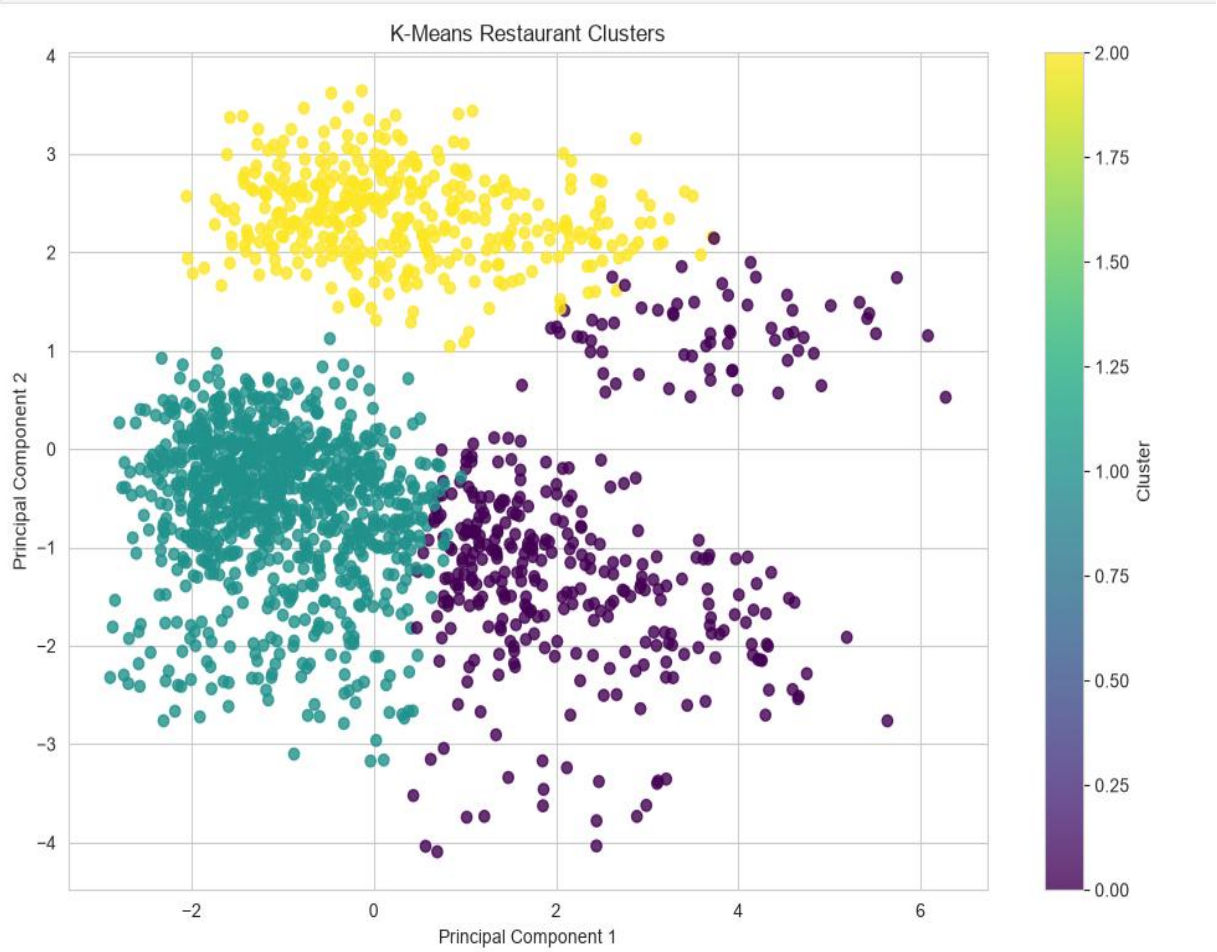


Figure 14: K-Means Cluster Analysis

5. Hierarchical Clustering

Hierarchical Clustering using the Ward linkage method was applied to create an alternative grouping of restaurants. The results were compared with K-Means to understand differences in cluster assignments.

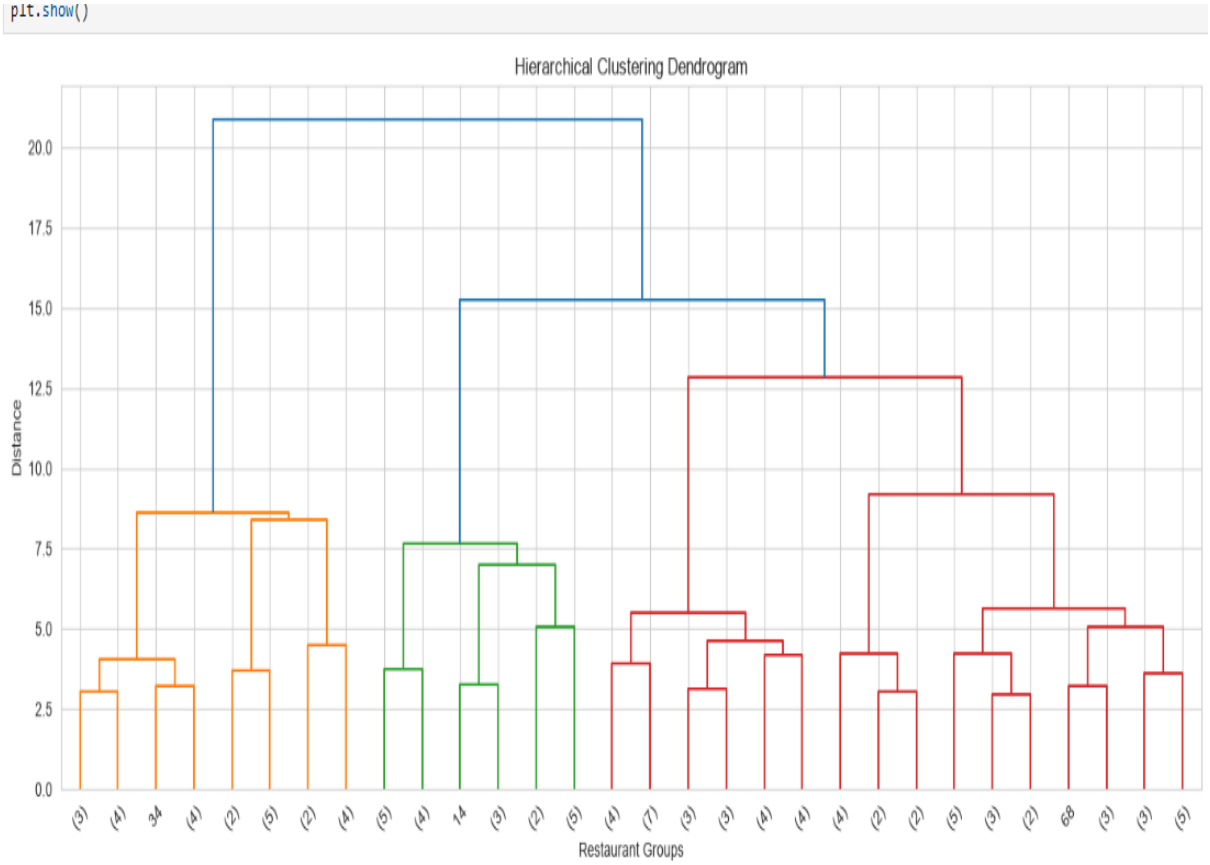


Figure 15: Hierarchical Clustering Results

6. DBSCAN Clustering

DBSCAN was explored as an additional density-based clustering technique. It can identify dense groups and observations that do not belong to a sufficiently dense region.

Its results were considered as an additional comparison rather than the primary strategic classification method.

7. Cluster Profiling

Cluster profiling was performed to understand the characteristics of each identified group.

The analysis considered indicators such as:

- Revenue and net profit.
- Monthly orders.
- Average order value.
- Cost burden.
- Growth factor.
- Delivery-channel contribution.
- Profit margin.

8. Strategic Restaurant Classification

Based on cluster profiles and business indicators, restaurants were interpreted using strategic categories.

Examples of the strategic labels considered in the project include:

- Expansion Ready.
- Operationally Strong but Under-Scaled.
- Aggregator Dependent and Margin Fragile.
- Profit Strong but Growth Constrained.
- Stabilize and Optimize.

The final dashboard displays the categories present in the exported classification dataset.

These labels are descriptive interpretations of cluster characteristics, not verified predictions of business success.

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)
```

	RestaurantName	KMeans_Cluster	StrategicCategory	GPI_Score
0	Urban Burgers House	1	Operationally Strong but Under-Scaled	53.020658
1	Urban Burgers Diner	0	Profit Strong but Growth Constrained	59.185078
2	King Burgers Eatery	0	Profit Strong but Growth Constrained	61.273911
3	Classic Burgers Tavern	1	Operationally Strong but Under-Scaled	39.840978
4	Lucky Burgers Bistro	1	Operationally Strong but Under-Scaled	48.953859
5	King Burgers Kitchen	1	Operationally Strong but Under-Scaled	47.771354
6	Urban Burgers Eatery	1	Operationally Strong but Under-Scaled	50.131074
7	Urban Burgers Tavern	1	Operationally Strong but Under-Scaled	56.104448
8	Classic Burgers Tavern	1	Operationally Strong but Under-Scaled	40.006782
9	Lucky Burgers Cafe	1	Operationally Strong but Under-Scaled	43.989962
10	Lucky Burgers Eatery	0	Profit Strong but Growth Constrained	52.346554
11	King Burgers Eatery	1	Operationally Strong but Under-Scaled	54.453831
12	King Burgers Bistro	1	Operationally Strong but Under-Scaled	66.325814
13	Urban Burgers Diner	0	Profit Strong but Growth Constrained	65.703068
14	Top Burgers Diner	0	Profit Strong but Growth Constrained	65.671151

Figure 16: Strategic Restaurant Classification

DASHBOARD DEVELOPMENT

An interactive business analytics dashboard was developed using Streamlit and Plotly.

The dashboard presents the results of EDA, feature engineering, GPI calculation, and machine learning-based classification in a user-friendly interface.

1. Dashboard Features

Executive Overview: Displays key performance indicators such as total restaurants, total revenue, total net profit, and average GPI.

Strategic Classification: Presents the distribution of restaurants across the available strategic categories.

Financial Performance: Compares revenue and net profit across restaurant groups.

Growth Opportunity Ranking: Displays restaurants ranked according to their GPI scores.

Cluster Analysis: Provides visual information about the restaurant clusters.

Growth Readiness Assessment: Presents the GPI-based classification.

Restaurant Profile Explorer: Allows users to inspect individual restaurant information.

Business Insights: Displays important observations and strategic recommendations.

Interactive Filters: Allows users to filter results using subregion, cuisine type, segment, strategic category, and K-Means cluster.

Data Download: Provides the option to download filtered restaurant records.

2. Dashboard Performance Summary

The completed dashboard displays the following key performance indicators:

KPI	Reported Value
Total Restaurants	1,696
Total Revenue	$77,739,306.80

Total Net Profit	$7,866,306.13
Average GPI Score	50.82

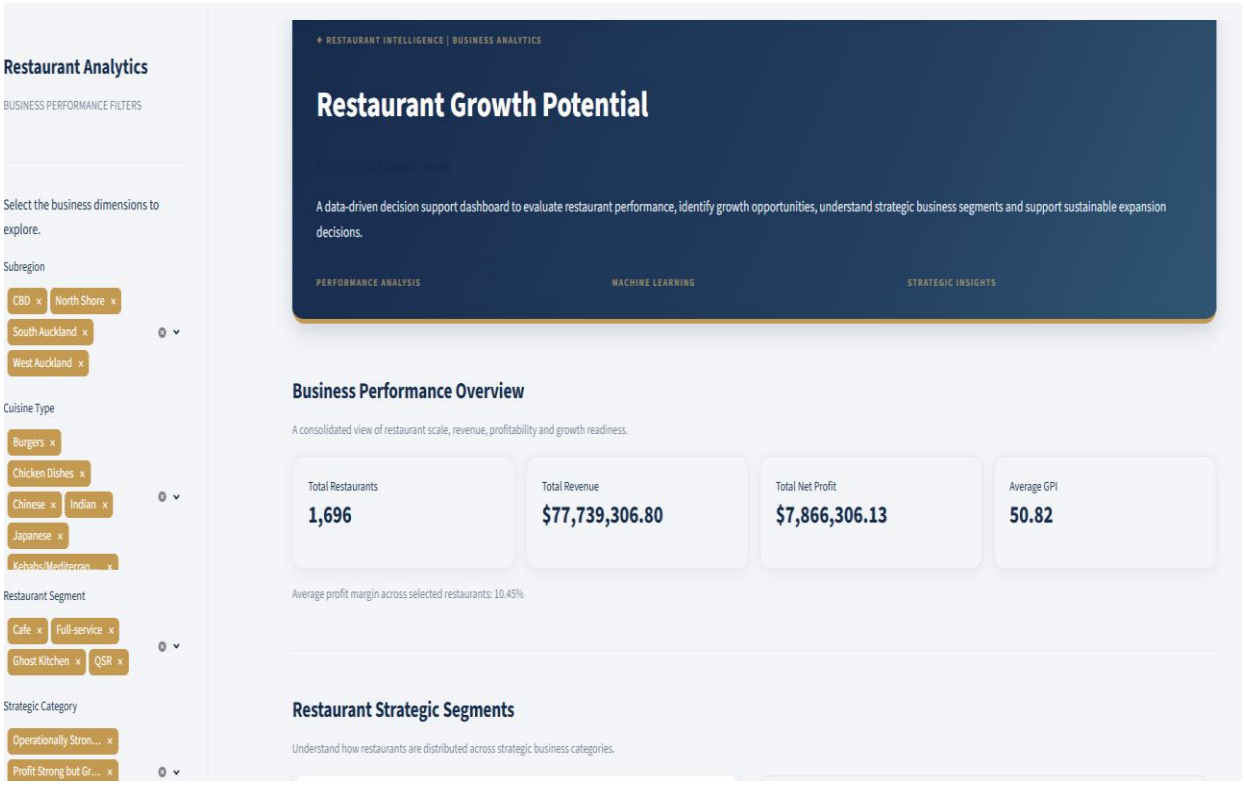


Figure 17: Streamlit Dashboard Overview

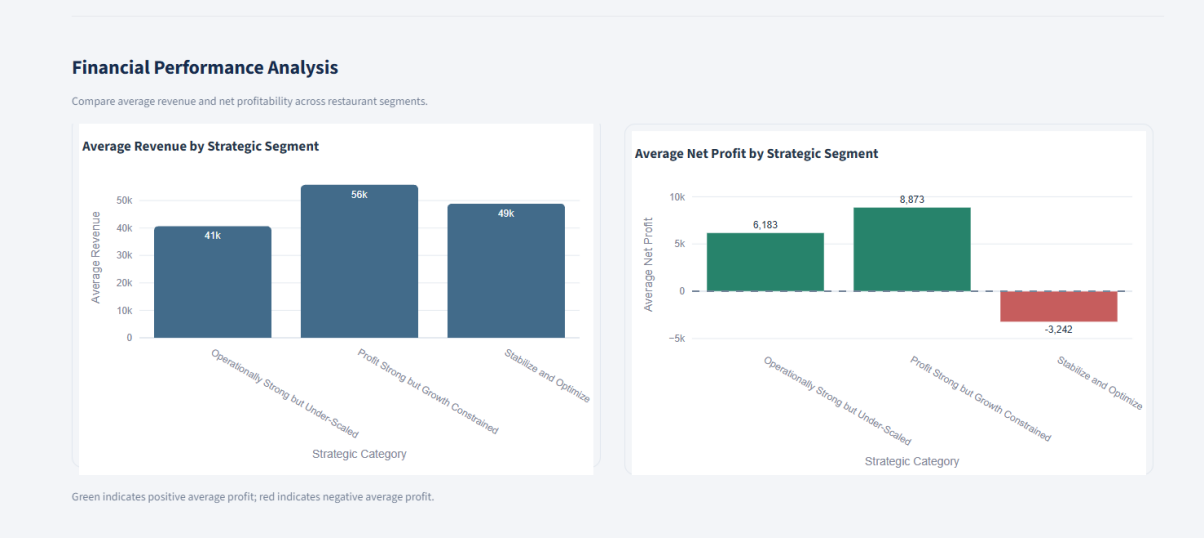


Figure 18: Financial Performance Dashboard

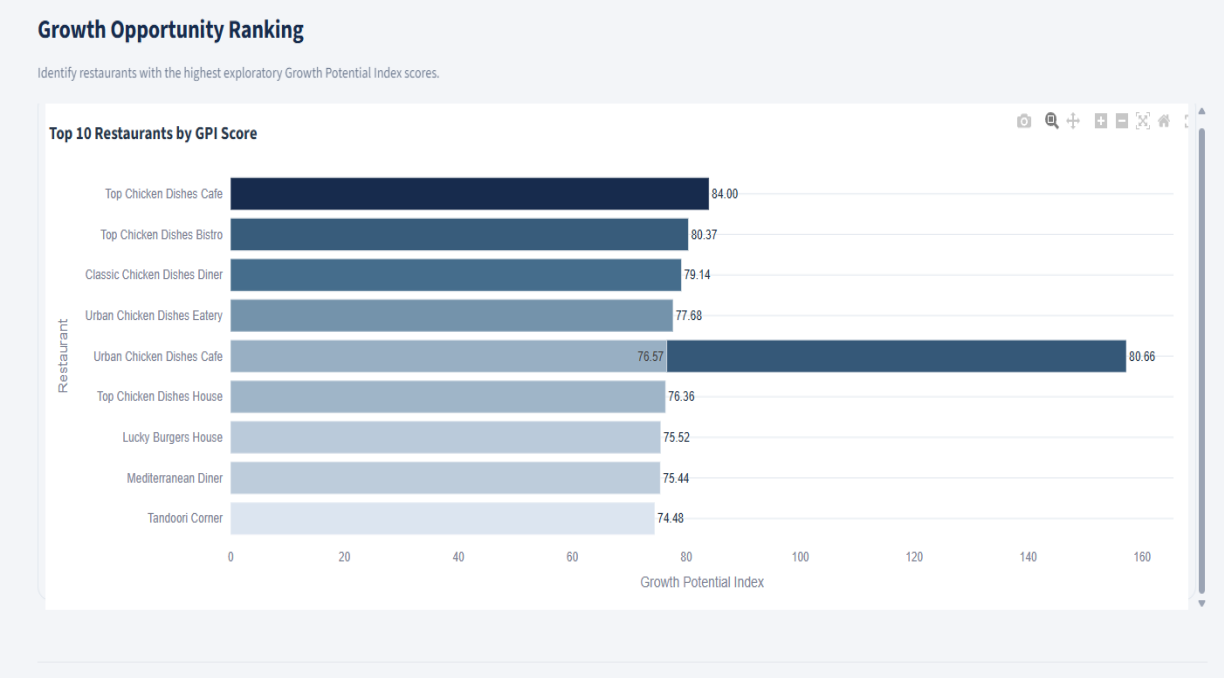


Figure 19: Growth Opportunity Ranking

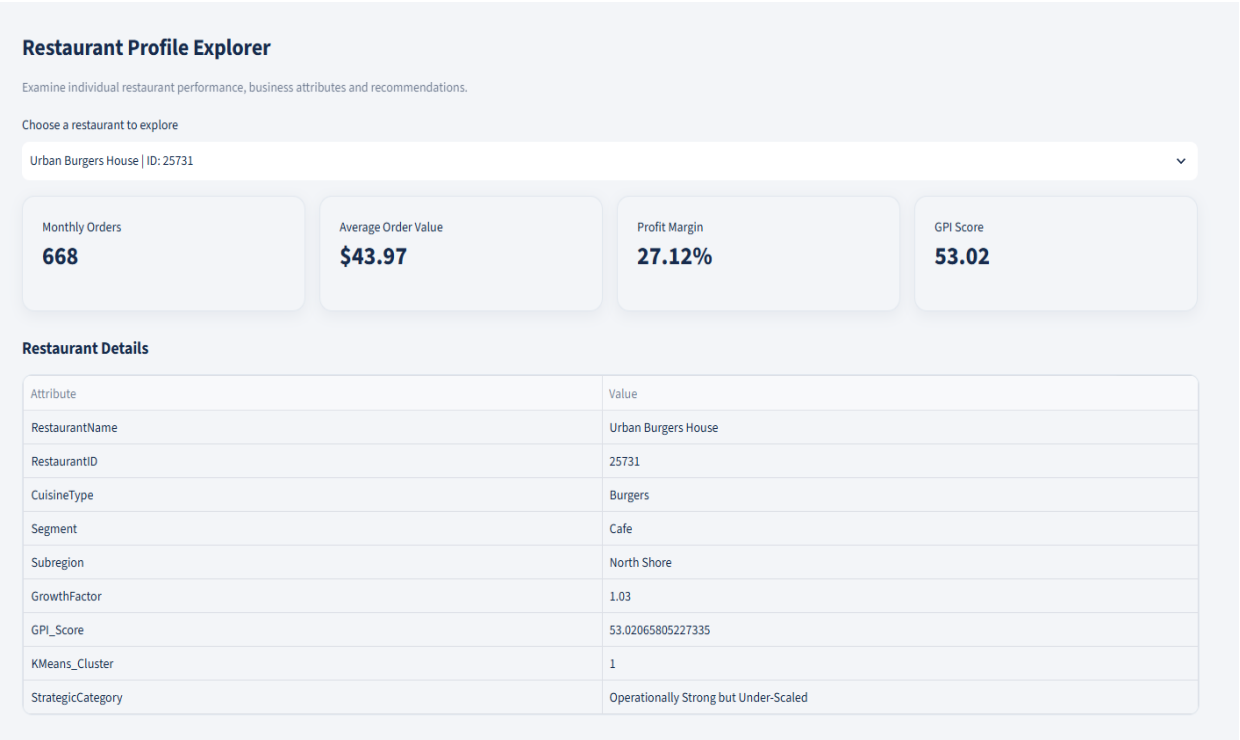


Figure 20: Restaurant Profile Explorer

Results & Findings

The project successfully combines exploratory data analysis, feature engineering, unsupervised clustering, and dashboard development to examine restaurant business performance.

The major findings are:

1. The final dashboard dataset contains 1,696 restaurant records.
2. The reported total revenue is $77,739,306.80.
3. The reported total net profit is $7,866,306.13.
4. The reported average Growth Potential Index is 50.82.
5. Revenue and profitability were examined across different sales channels.
6. Cost, commission, and delivery indicators were included in the analysis.
7. K-Means and Hierarchical Clustering were applied to explore groups of similar restaurants.
8. PCA was used for dimensionality reduction and visualization.
9. The dashboard presents the strategic categories found in the final processed dataset.
10. Interactive filters and visualizations make it easier to examine restaurant-level performance.

The findings provide an exploratory view of restaurant performance. They do not establish that a restaurant will achieve a particular level of future growth.

Business Insights

1. **Revenue and Profitability:** Revenue should be considered together with net profit because high sales do not necessarily indicate strong profitability.
2. **Sales Channel Performance:** Comparing in-store and delivery channels helps understand how restaurants generate revenue through different business channels.
3. **Customer Demand:** Monthly orders and average order value provide complementary information about restaurant demand and transaction value.
4. **Cost Management:** Operating expenses, commission rates, and delivery costs are important factors when examining financial performance.
5. **Delivery Operations:** Delivery radius, delivery orders, and delivery costs can help identify differences in delivery operations.
6. **Restaurant Segmentation:** Clustering helps organize restaurants into groups with similar business characteristics.
7. **Growth Assessment:** The GPI provides a single exploratory score for comparing selected business indicators.
8. **Decision Support:** The dashboard provides a centralized view of restaurant performance and helps users explore business information.

STRATEGIC RECOMMENDATIONS

Based on the analysis, the following recommendations can be considered:

1. Evaluate revenue and net profit together before making expansion decisions.
2. Review commission rates and channel-level profitability before increasing dependence on third-party delivery platforms.
3. Identify restaurants with strong operational indicators but comparatively limited scale for further growth evaluation.
4. Review operating costs for restaurants requiring stabilization or improvement.
5. Evaluate delivery demand and delivery costs before expanding delivery coverage.
6. Use individual restaurant profiles along with cluster-level summaries when making decisions.
7. Review the GPI indicators and weights with business experts before using the score for formal investment decisions.
8. Update the dashboard periodically with validated business data.

Conclusion

The project, “Restaurant Growth Potential Modeling & Strategic Classification System,” provides an analytical approach for evaluating restaurant business performance using the SkyCity Auckland Restaurants & Bars dataset.

Exploratory Data Analysis was performed to understand restaurant demand, revenue, profitability, operating costs, and delivery-channel performance. Feature engineering was used to create additional business indicators, followed by the development of an exploratory Growth Potential Index.

Unsupervised machine learning techniques, including K-Means and Hierarchical Clustering, were applied to identify groups of restaurants with similar characteristics. PCA supported dimensionality reduction and cluster visualization. The resulting cluster profiles were interpreted to develop strategic restaurant categories.

An interactive Streamlit dashboard was also developed to present key performance indicators, financial comparisons, restaurant rankings, cluster information, and business insights.

Overall, the project demonstrates how data analytics and machine learning can support structured business performance analysis. The GPI and strategic categories provide exploratory decision-support information and should be validated against business expertise and future outcomes before being used as predictive measures.

Future Scope

The project can be enhanced in several ways:

1. Validate the Growth Potential Index using historical restaurant performance.
2. Refine GPI weights using expert feedback and business outcomes.
3. Explore additional clustering algorithms and compare cluster stability.
4. Develop supervised machine learning models if reliable historical growth labels become available.
5. Integrate updated restaurant data for continuous performance monitoring.
6. Add geographic visualization to compare restaurants across subregions.
7. Introduce automated PDF and Excel reporting.
8. Add more detailed restaurant-level recommendations.
9. Improve dashboard access control and data security.
10. Extend the system to support periodic business performance tracking.

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